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# State of AI in the Enterprise

## The untapped edge

January 2026

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# Introduction

Throughout history, some of our greatest leaps forward have come when human ingenuity combines with transformative technology. Each era's boldest breakthroughs—from harnessing steam to building the internet—began when people leveraged new tools to amplify their potential. Artificial intelligence (AI) is the latest chapter in this story. It has already transformed the way we work and create, yet we have barely scratched the surface of what's possible when human expertise and AI capabilities unite.

Business leaders today face an unprecedented challenge: moving beyond pilots to truly integrating AI into the heart of their organizations. It requires a deliberate shift in which people set a vision and make responsible choices, and AI provides the insights, speed, and scale to deliver against that ambition. That means redesigning core processes and operating models with AI, ensuring that human strengths—such as judgment, creativity, empathy, and relationship-building—are elevated, not automated.

This year's State of AI in the Enterprise report captures this pivotal moment. On the one hand, we see clear acceleration from organizations: with wider workforce access to AI tools, early productivity gains, and growing confidence in AI's potential. On the other hand, we see a gap between experimentation and true enterprise transformation. Many organizations are primarily using AI to drive efficiency, while a smaller group are pulling ahead by beginning to reimagine business models, offerings, roles, and ways of working. Leading organizations are rethinking how work gets done, how teams are structured, and how human and machine capabilities can complement one another in everyday operations.

Key trends are also reshaping the future of AI: Sovereign AI is redefining technological autonomy for nations and organizations alike, impacting trust and competitiveness; Agentic AI is enabling autonomous reasoning and action, raising new governance and control challenges; Physical AI is merging the digital and material worlds, making safety and human oversight critical. These trends signal a future that is even more transformative, demanding organizational strategies that align technological scale with human guidance and intent.

Now is the time to be visionary—reshaping organizational structures, roles, and workflows in ways that may not have been possible before. We hope that the insights in this report inspire leaders to take practical steps forward, while always keeping people, purpose, and trust at the center.

Together, we have the opportunity and the responsibility to fundamentally rethink how humans and AI work together to transform the future of work and create long-term value.

# Overview

Organizations today stand at the untapped edge of AI's potential. Ongoing developments in agentic, physical, and sovereign AI present new challenges and opportunities. Momentum is building, yet the greatest gains still lie ahead as organizations translate early progress into scalable impact. As AI expands beyond the digital core, success will ultimately hinge on a company's ability to move from ambition to activation—turning experiments and potential into real-world business value at scale.

Deloitte's latest *State of AI in the Enterprise* survey captured insights from more than 3,200 business and IT leaders around the world with direct involvement in their companies' AI initiatives. This report explores the survey insights in detail, sharing critical actions for leaders to consider as they continue on their journey with AI.

**Key findings**

## AI is moving from the pilot and experimentation phase to enterprise scaling as worker access to AI expands

Surveyed companies have broadened worker access to AI by 50% in just one year—growing from fewer than 40% to around 60% of workers now equipped with sanctioned AI tools. While only 25% of respondents said their organization has moved 40% or more of their AI experiments into production to date, 54% expect to reach that level in the next three to six months.

## AI transformation reveals productivity for most, business reimagination for a few

AI is already boosting productivity and efficiency; just a subset are using it to rewrite the business. Today, 34% of companies are starting to use AI to deeply transform their businesses, 30% are redesigning key processes around AI and the remaining 37% are only using AI at a surface level with little or no change to underlying business processes.* While each are capturing productivity and efficiency gains, just the first group are truly reimagining their businesses rather than optimizing what already exists.

* Figures may not sum to 100% due to rounding.

## Companies are focused on building AI fluency instead of redesigning work around AI

Despite high expectations for automation, 84% of companies have not redesigned jobs or the nature of work itself around AI capabilities. Insufficient worker skills are seen as the biggest barrier to integrating AI into the business, but fewer than half of companies are making significant adjustments to their talent strategies. Most are focused on educating employees, but far fewer are rearchitecting roles, workflows, and career paths.

## With sovereign AI taking hold, where technology is built matters as much as what it can do

Sovereign AI is about more than technology ownership. It's about strategic independence. More than 3 in 4 companies (77%) say the location of AI development is a key factor when choosing new technologies, signaling that geographic sovereignty is now as important as innovation.

**77%** **of surveyed companies say the location of AI development is a key factor when choosing new technologies.**



## AI agents are scaling faster than the guardrails

Autonomous AI agents are racing into the enterprise, but oversight is lagging. Nearly 3 in 4 (74%) companies plan to deploy agentic AI within two years. Yet, 1 in 5 (21%) report having a mature model for governance of autonomous agents, raising the specter of unintended risks.

## Physical AI is already embedded in operations—and its footprint is growing fast

Physical AI is rapidly becoming integral to operations worldwide, with 58% of companies already using it to some extent and adoption projected to hit 80% within two years. While manufacturing, logistics, and defense lead the way globally, markets in Asia Pacific are leading adoption, driving widespread integration of robotics, autonomous vehicles, and drones—setting the pace for the next wave of industrial automation.

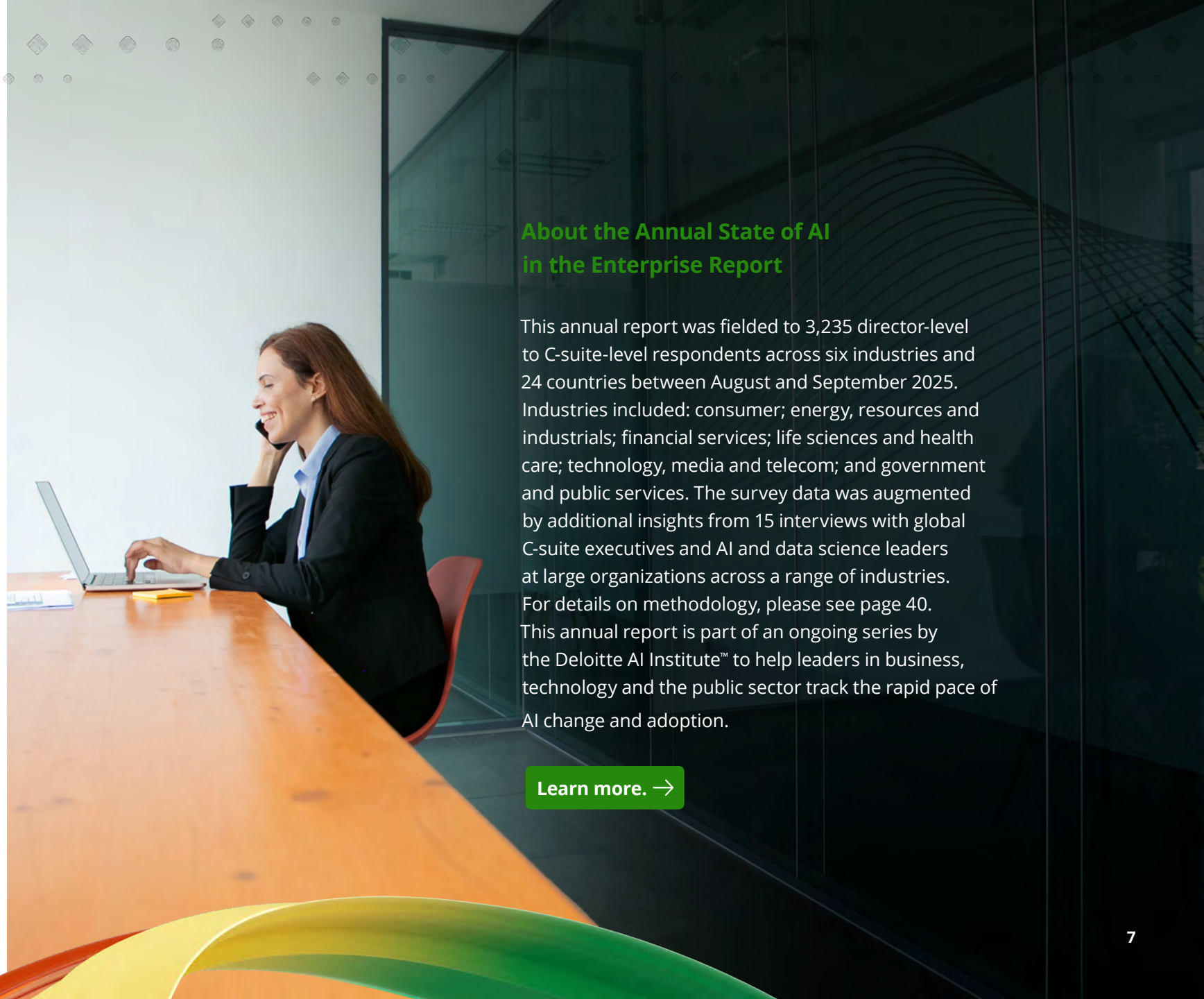
**74%** **Nearly 3 in 4 companies plan to deploy agentic AI within two years.**



# Leaders feel more strategically ready for AI than operationally ready in infrastructure and talent.

Despite the rapid evolution of AI beyond Generative AI (GenAI) to agentic and physical AI, 42% of companies believe their strategy is highly prepared for AI adoption and 30% say the same about risk and governance—both rising since last year. Perceptions of preparedness shifted down slightly for technical infrastructure, data management and talent, revealing the persistent challenge of modernizing systems and skills at the speed of innovation.

**42%** **of companies believe their strategy is highly prepared for AI adoption.**



## About the Annual State of AI in the Enterprise Report

This annual report was fielded to 3,235 director-level to C-suite-level respondents across six industries and 24 countries between August and September 2025. Industries included: consumer; energy, resources and industrials; financial services; life sciences and health care; technology, media and telecom; and government and public services. The survey data was augmented by additional insights from 15 interviews with global C-suite executives and AI and data science leaders at large organizations across a range of industries. For details on methodology, please see page 40.

This annual report is part of an ongoing series by the Deloitte AI Institute™ to help leaders in business, technology and the public sector track the rapid pace of AI change and adoption.

**Learn more.** →

# Key findings

## AI is moving from the pilot and experimentation phase to enterprise scaling as worker access to AI expands

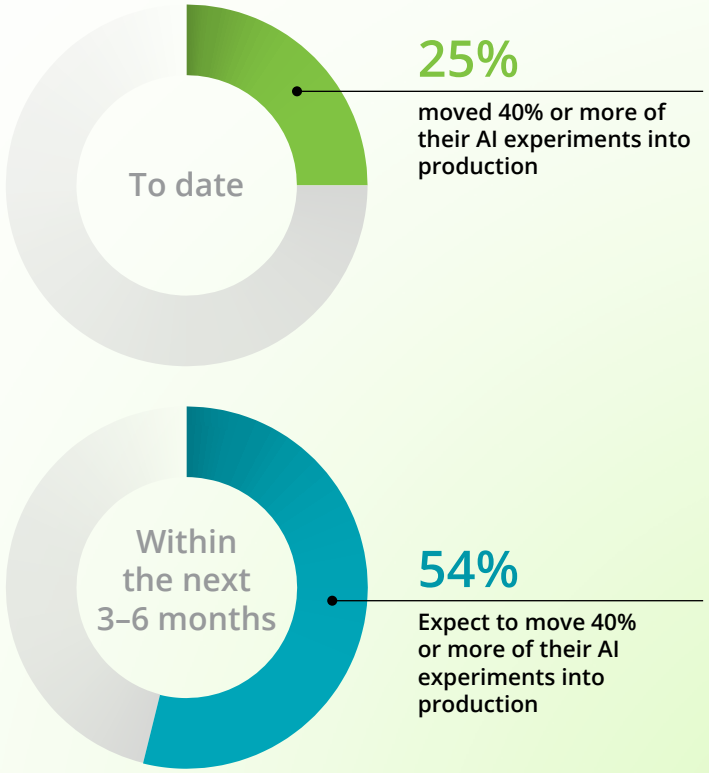
According to our latest survey, workforce access to AI has expanded by 50% in just one year—growing from under 40% to under 60% of workers with sanctioned access to AI tools, and 11% of leading companies currently provide workers with near-universal (more than 80%) access to sanctioned AI tools. However, among those workers with access, fewer than 60% use it in their daily workflow—a pattern that remains largely unchanged from last year. This suggests that while access is widening, enterprise AI remains underutilized, and its productivity and innovation potential are still largely untapped.

**The scale acceleration is beginning**

Moving from pilot to production is arguably the most important step in capturing AI value—yet this is where many companies stall. While enterprises are experimenting with AI at an accelerating pace, many struggle to scale these experiments into solutions that deliver measurable business impact.

Today, 25% of respondents said their organization has moved 40% or more of their AI experiments into production to-date (figure 1); however, 54% expect to reach that level in the next three to six months, demonstrating the pathway to value is clear and achievable. These early scalers highlight an accelerated shift from pilots to enterprise scale.

**Figure 1: Proportion of AI experiments deployed** *(percent)*




Question: In your estimation, what percentage of your AI experiments (e.g., pilots, test cases, etc.) have been deployed to date into your organization (moved into production)?
N=3,235

## The proof-of-concept trap

Why do so many pilots fail to reach production? The answer lies in a fundamental mismatch between pilot and production requirements. A pilot typically can run with a small team in a few months using cleansed data in an isolated environment. However, production deployment typically requires infrastructure investment, integration with existing systems, security reviews, compliance checks, monitoring systems, and ongoing maintenance—each of which demand significantly more resources and coordination.

Production also exposes realities that pilots can hide. Models that achieve high accuracy in testing may prove inadequate when handling edge cases at scale. Use cases estimated to take three months can stretch to 18 months when integration complexities emerge. Failures that were learning opportunities in pilots become business risks in production.

Organizations that experiment with AI often see positive results in controlled conditions but cannot consistently predict which use cases will yield the highest return on investment. This lack of clear value realization creates a vicious cycle where companies continue funding new pilots—which are relatively low cost and lower risk—rather than facing the harder work of scaling up existing successes.

The challenge of moving beyond isolated pilots to real enterprise impact is echoed by the people leading AI efforts on the ground. According to one healthcare AI leader: "If there is no coherent AI strategy in organizations, you are likely to see pilot fatigue. You're chasing the next shiny object, pressured to do something with AI without a real plan. I've seen many instances where people embark on pilots, but when asked how they'll scale up if successful, they often don't have an answer. Without a clear roadmap, executing a hundred pilots just leads to poor results and failed value creation."

**"If there is no coherent AI strategy in organizations, you are likely to see pilot fatigue."**

# AI transformation reveals productivity for most, business reimagination for a few

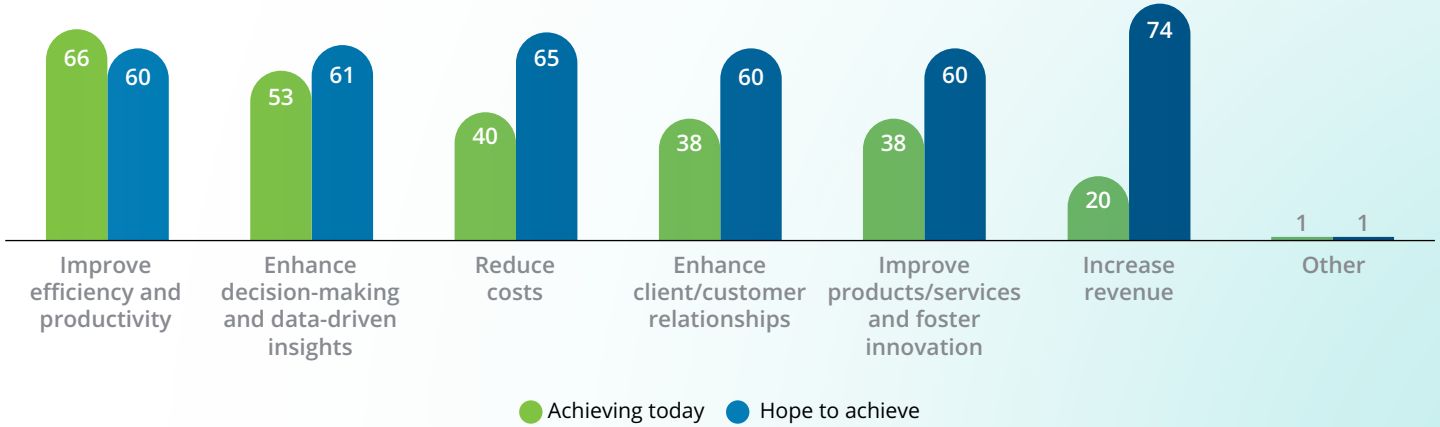
AI's real-world business impact is rising fast, with 25% of leaders now reporting that AI is having a transformative effect on their companies—more than double from 12% a year ago. Trust and investment are also surging, with 84% of organizations increasing their AI investments and 78% of leaders reporting greater confidence in the technology. Yet, most companies are only at the edge of large-scale AI-driven transformation.

## Beyond productivity and efficiency

AI is already delivering widespread gains in efficiency and productivity; however, benefits in other areas are taking longer to achieve. In particular, revenue growth largely remains an aspiration, with 74% of organizations hoping to grow revenue through their AI initiatives in the future compared to just 20% that are already doing so (figure 2).

These numbers suggest AI is on the verge of breaking out and delivering a wide range of benefits that go far beyond efficiency and productivity improvements. Ultimately, however, success with AI isn't just about boosting efficiency or even growing revenue. It's about achieving strategic differentiation and a lasting competitive edge in the marketplace.

**Figure 2: AI benefits achieving today vs. hope to achieve** *(percent)*



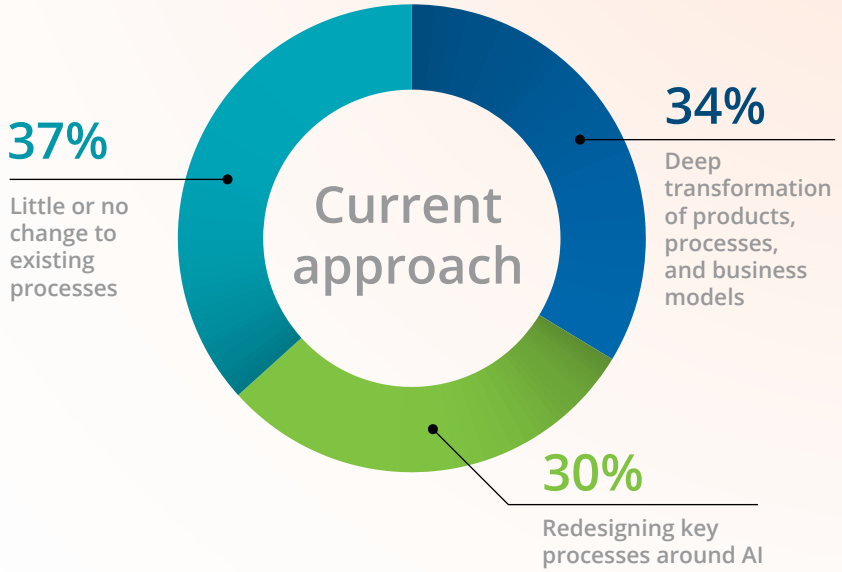

Question: With regards to benefits from your AI efforts: Which benefits are you achieving today? Which benefits do you hope to achieve?
N=3,235

**Organizations are redefining how they work, but not all are diving to the same depth**

Among the surveyed companies, one-third (34%) are already starting to use AI to *deeply transform* their businesses—creating new products and services, reinventing core processes, or even fundamentally changing their business models. Another third (30%) are *redesigning key processes* around AI but keeping their business models intact. And the remaining third (37%) are using AI at a more *surface level*, with little or no change to existing processes* (figure 3). While each are capturing productivity and efficiency gains, only the first group are truly reimagining their businesses rather than optimizing what already exists.

The head of AI and automation and global engineering at a mining company explained how they took a bold, strategic approach to AI transformation by embedding AI into the company's core offerings and products, most notably transforming traditional mining equipment into intelligent, connected platforms with sensors and predictive analytics. "AI is much more than a technology...we wanted to give it to everyone for everyday usage and make it pervasive everywhere. But we also wanted to disrupt the market." Rather than focusing solely on internal improvements, the organization is reimagining its business by using AI to unlock new value for both itself and its clients, and to create new digital solutions and revenue streams.

Figure 3: Current approach to transformation with AI*




Question: Which best describes your organization's current approach to process transformation during AI implementation?
* Figures may not sum to 100% due to rounding.
N=3,325

**"AI is much more than a technology...we wanted to give it to everyone for everyday usage and make it pervasive everywhere. But we also wanted to disrupt the market."**

* Figures may not sum to 100% due to rounding.

## Companies are focused on building AI fluency instead of redesigning work around AI

Within a year, more than a third of surveyed companies (36%) expect at least 10% of their jobs to be fully automated. The majority of surveyed companies (82%) expect at least 10% of their jobs to be fully automated when looking out three years.

These changes require careful thinking about career pathways. Leaders in the qualitative interviews expressed concerns about potential disruption to professional development pipelines as a result of automation. Entry-level jobs involving data entry, reconciliation, and first-level customer support at their companies are being prioritized for automation, but these jobs are often the starting point for longer careers. Organizations will likely need to develop alternate pathways for professional advancement, ensuring that employees have expertise that includes foundational processes.



**36%** **of surveyed companies expect at least 10% of their jobs to be fully automated within a year.**

## Most companies have yet to redesign jobs around AI

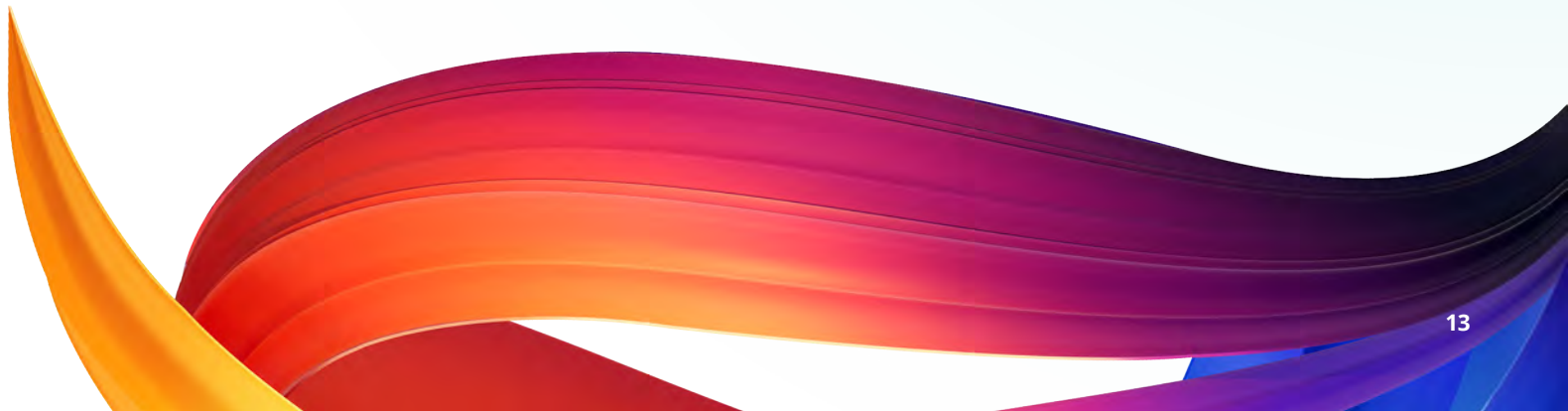
Despite high expectations for automation, 84% of companies have not redesigned jobs around AI capabilities.

AI doesn't just augment existing processes. It often requires fundamentally rethinking operating models and how work gets done.[1] A loan officer who has always used judgment and experience to approve loans must now work with an AI system that provides recommendations. This raises questions: *When should they override the AI? How do they explain decisions to customers? What happens to their expertise and career trajectory?*

Entry-level and task-aligned roles could be most affected, as automation may replace common, time-consuming tasks. However, as front-line jobs become more automated, supervisor and managerial roles will likely shift toward orchestration of human-AI teams. This is prompting many organizations to explore flatter structures: 53% have considered pod-based or non-hierarchical models since fewer roles require supervision of large teams; however, only 16% have moved to such models to a great or maximum extent.

**84%**

**of surveyed companies have not redesigned jobs around AI capabilities.**



## Talent strategies are falling short

Worker sentiment about AI is mixed but cautiously positive. While 13% of non-technical workers are highly enthusiastic about AI and are proactively seeking to use it—and 55% are at least open to exploring it—skepticism remains: 21% prefer not to use AI but will do so if required, and 4% actively distrust and avoid it.

According to the leaders surveyed, insufficient worker skills are the biggest barrier to integrating AI into existing workflows. Yet, fewer than half of companies are making significant adjustments to their talent strategies, with most (53%) simply focusing on educating employees to raise AI fluency (figure 4). While most are focused on educating employees, far fewer are rearchitecting roles, workflows, and career paths.

A director of AI and innovation at a major logistics organization emphasized the importance of workforce upskilling and supporting business teams as they adapt to new AI technologies: "We are reskilling our people on the business side—investing a lot to ensure they adopt the new AI tools so they can deliver bigger, better, and smarter." The company's vision for workforce transformation isn't just limited to basic upskilling. Rather, the focus is on making sure employees can move from traditional roles into more strategic positions—supported by AI tools. "For example, in the future we would like to see AI enable today's pricing analysts to become pricing strategists."

**Figure 4: Talent strategy adjustments due to AI** *(percent)*




Question: How is your organization adjusting its talent strategies because of the adoption of AI tools/capabilities?
N=3,235

# With sovereign AI taking hold, *where* technology is built matters as much as *what* it can do

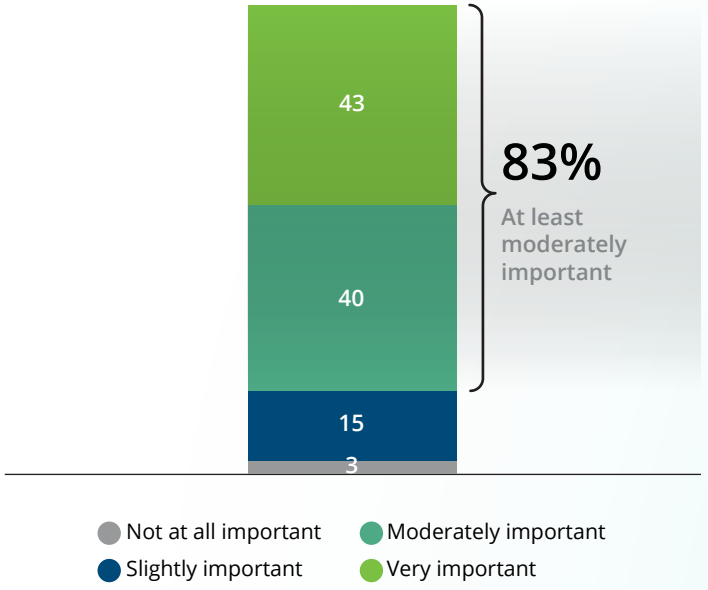
Sovereign AI is when a country—and the companies operating within it—design, train, and deploy AI under their own laws, on infrastructure they control, using locally governed data. The goal is to reduce dependence on foreign vendors for critical AI capabilities.

## A new boardroom issue

More than 8 in 10 companies (83%) view sovereign AI as at least moderately important to their strategic planning, and nearly half (43%) rate it as very important or extremely important (figure 5).

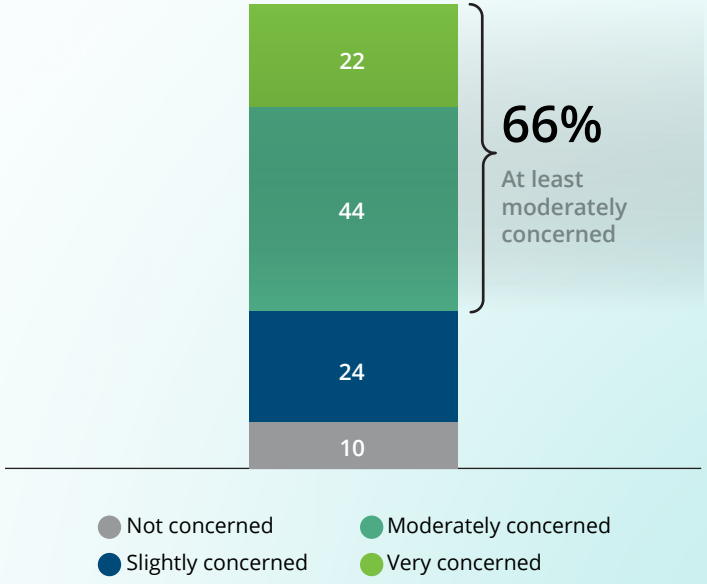
Similarly, 66% of companies express at least moderate concern about reliance on foreign-owned AI technologies and infrastructure, with 22% very concerned or extremely concerned (figure 6).

**Figure 5: Importance of data residency constraints and compute considerations in strategic planning*** *(percent)*




Question: How important are data residency constraints and in country/region compute considerations to your organization's strategic planning?
* Figures may not sum to 100% due to rounding.
N=3,235

**Figure 6: Level of concern over reliance on foreign-owned AI** *(percent)*




Question: To what extent are you concerned about your organization's reliance on foreign-owned AI technologies and compute provision?
N=3,235

## Sovereign AI in practice

The rise of sovereign AI has immediate practical implications. Companies working across borders must navigate complex requirements that vary by country, forcing them to build customized solutions for different markets. More than 3 in 4 companies (77%) now factor an AI solution's country of origin into their vendor selection decisions, and nearly 3 in 5 (58%) now build their AI stacks primarily with local vendors. This signals that geographic sovereignty is now as important as innovation.

As the former vice president of observability at a major telecommunications company noted, "I've been working with a lot of international companies lately that are adamant we use an in-country infrastructure. We're taking the approach with some of those customers to do a distilled small language model for them that meets the import/export control rules because we build it in their country. With state-run companies in particular, there is skepticism when you're using something from outside the country."

Sovereign AI pressures vary significantly by geography and industry. Only 11% of companies in the Americas rely on foreign-sourced solutions for the majority of their AI stack, compared to 32% of Europe/Middle East/Africa (EMEA) companies. In the US, state-level deployment considerations are increasingly shaping the landscape.

Ultimately, sovereign AI isn't just about technology ownership. It's about strategic independence. By building on infrastructure within its own control (fueled by its own data, models, talent, and ecosystem) a company has the ability to innovate securely and responsibly.

**77%**

**of surveyed companies now factor an AI solution's country of origin into their vendor selection decisions.**



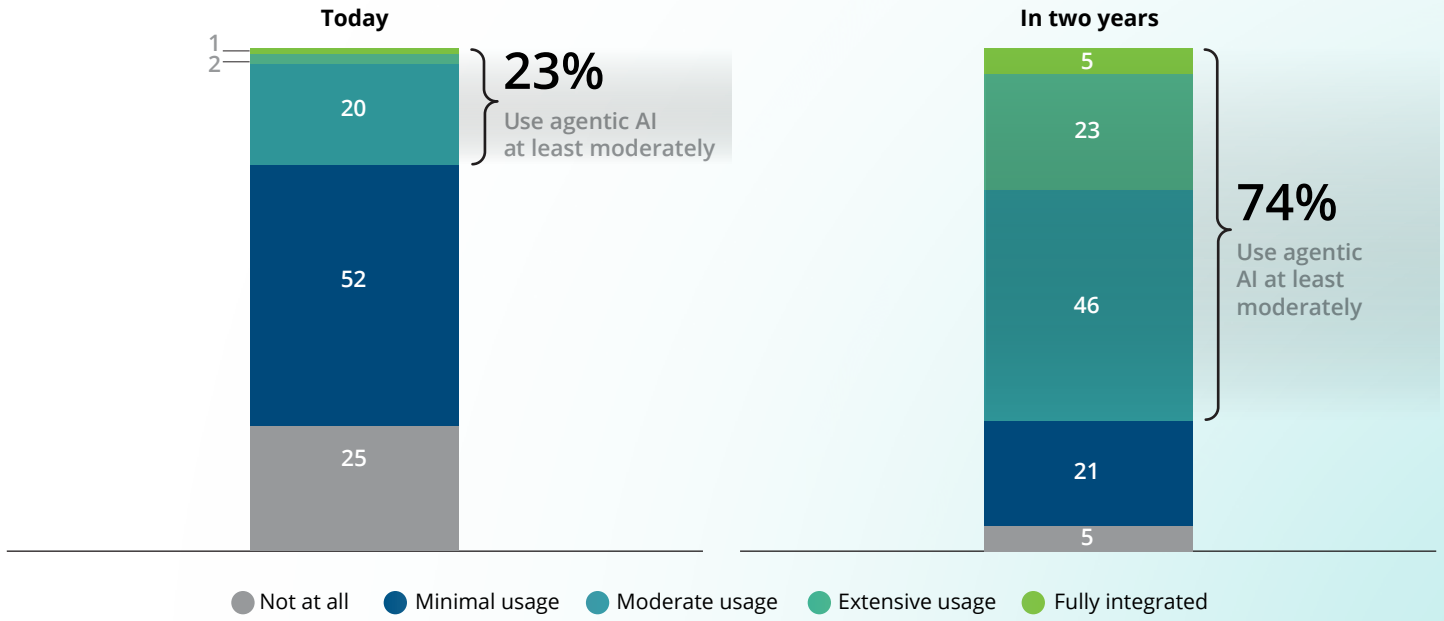
# AI agents are scaling faster than the guardrails

After years of non-AI chatbots that answered basic questions, companies are now deploying sophisticated AI agents that can set goals, reason through multi-step tasks, use tools and application programming interfaces (APIs), and coordinate work with people or other agents. This shift transforms AI from a source of information and insights into a system that could perform in a different capacity.

## Agentic AI will surge

In last year's State of AI report, 26% of respondents said their organizations were already exploring autonomous agent development to a large or very large extent. Those early exploration efforts are now starting to translate into real-world use and are poised to drive the sharp rise in adoption expected in the near term. Today, 23% of companies are using agentic AI at least moderately. However, within the next two years agentic AI is expected to become nearly ubiquitous, with nearly 3 in 4 companies (74%) using it at least moderately, 23% using it extensively, and 5% fully integrating it as a core component of their operations (figure 7).

**Figure 7: Extent of agentic AI usage** *(percent)*




Question: To what extent is your organization utilizing agentic AI in its operations?
N=3,235

## AI agents unlock a vast array of use cases

While agentic AI is expected to have the highest impact in customer support, use cases for supply chain management, R&D, knowledge management, and cybersecurity are also seen as having particularly high potential.[2] The enterprises we interviewed are already deploying AI agents across multiple functions:



**A financial services company** is building agentic workflows to automatically capture meeting actions from video conferences, draft communications to remind participants of their next steps, and track follow-through.



**An air carrier** is using AI agents to help customers complete the most common transactions, such as rebooking a flight or rerouting bags, freeing up time for human agents to address more complex matters.



**A manufacturer** is using AI agents to support new product development initiatives, leveraging AI to find the optimal balance between competing objectives such as cost and time-to-market.



**In the public sector**, AI agents are being used to cover workforce shortages, partnering with human workers to complete key processes.

**85%** **of companies expect to customize agents to fit the unique needs of their business.**

## AI agents don't eliminate the value of humans

Adoption of agentic AI may increase the need for uniquely human strengths, such as adaptivity and judgment, in the near term. "We thought we were going to automate jobs," said the former VP of observability at a major telecommunications company. "The truth is, you're not. You're going to give existing workers force multipliers where they can be more effective. Maybe someday these things will start to become headless where they just feed off a dashboard metric and you can pull back staff to wait on an alert that wakes somebody up or flashes red on the screen if something really bad happens. But initially it is going to be more work for those people. They're not going to be cooling their heels; they're going to be watching these agents, making sure the volume metrics are right, making sure the qualitative metrics are right, and being there to interact with them if they hit a human-in-the-loop gate and need to interact with a human for accountability purposes."

**"We thought we were going to automate jobs. The truth is, you're not. You're going to give existing workers force multipliers where they can be more effective."**



## Managing the risks of agentic AI

About 1 in 5 (21%) companies surveyed report currently having a mature model for governance of autonomous agents. Given the technology's rapid adoption trajectory, this could be a significant limitation. As agentic AI scales from pilots to production deployments, establishing robust governance should be essential to capturing value while managing risk.

Governing agentic AI requires new approaches beyond traditional AI oversight. Unlike conventional AI systems that provide recommendations for humans to act upon, agents take actions directly—making purchases, sending communications, or modifying systems. Organizations need to establish clear boundaries for agent autonomy, defining which decisions agents can make independently versus which require human approval. Real-time monitoring systems that track agent behavior and flag anomalies are essential, as are audit trails that capture the full chain of agent actions to help ensure accountability and enable continuous improvement.

Companies seeing the most success are taking a measured approach—starting with lower-risk use cases, building governance capabilities, and scaling deliberately. This includes cross-functional governance structures that bring together IT, legal, compliance, and business unit leaders to set policies, monitor performance, and manage escalations. Rushing to deploy agents widely before establishing these governance foundations can expose organizations to significant risks.



**21%**

**of companies report currently having a mature model for governance of autonomous agents.**

A closer look at governance

# AI governance as the catalyst for growth

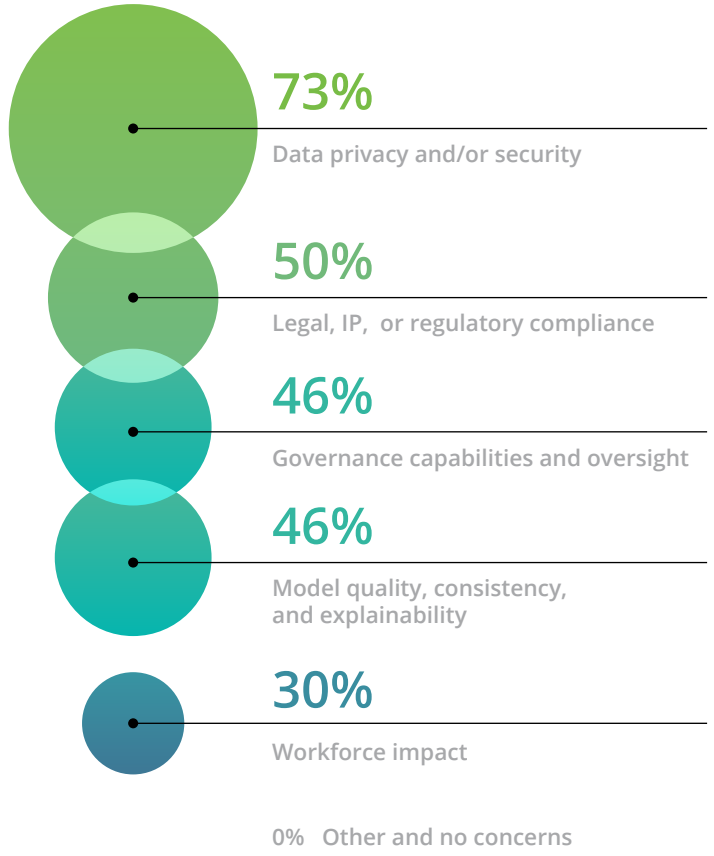
As AI moves from experimentation to deployment, governance is the difference between scaling successfully and stalling out. In the AI era, governance is more than guardrails—it's the catalyst for responsible growth.

The AI risks companies are most worried about all relate to governance. Data privacy and security tops the list at 73%, followed by legal, intellectual property, and regulatory compliance (50%), governance capabilities and oversight (46%), and model quality, consistency, and explainability (46%) (figure 8).

Insights gathered during qualitative interviews with AI leaders reveal that governance is a concern. In some organizations, leaders are learning AI models have been deployed into production without formal oversight or monitoring processes. In particular, one AI leader discovered that there wasn't a clear inventory of all the AI tools and models currently active because development efforts occurred without systematic tracking or centralized visibility over what is running in production.

Organizations that are serious about capturing AI value should treat governance as a strategic capability, not an afterthought. Those that build strong governance frameworks now will be positioned to scale AI quickly and safely. On the other hand, those that treat governance as a checkbox exercise may find themselves unable to move AI from pilot to production, held back by the very risks they failed to address.

**Figure 8: AI risks most concerning** *(percent)*



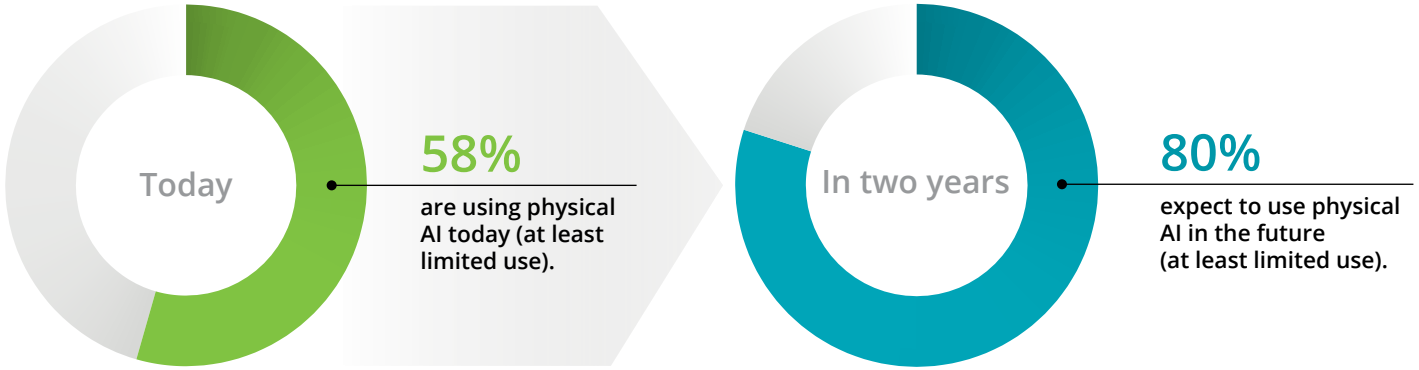

Question: Which of the following risks related to AI tools / applications is your organization most concerned about?
N=3,199

# Physical AI is already embedded in operations—and its footprint is growing fast

Physical AI is the class of AI systems that perceive the real world, make decisions, and drive physical actions through machines or control systems. It sits at the intersection of AI and machine learning, sensors, controls, and robotics.

Physical AI integration is already expanding, with 58% of companies reporting at least limited use of physical AI, and among these, 18% are leveraging it to a moderate or greater extent. However, the percentage of companies using physical AI in any capacity is expected to reach 80% within two years—with 15% using physical AI extensively and 3% fully integrating it as a core element of their operations (figure 9).

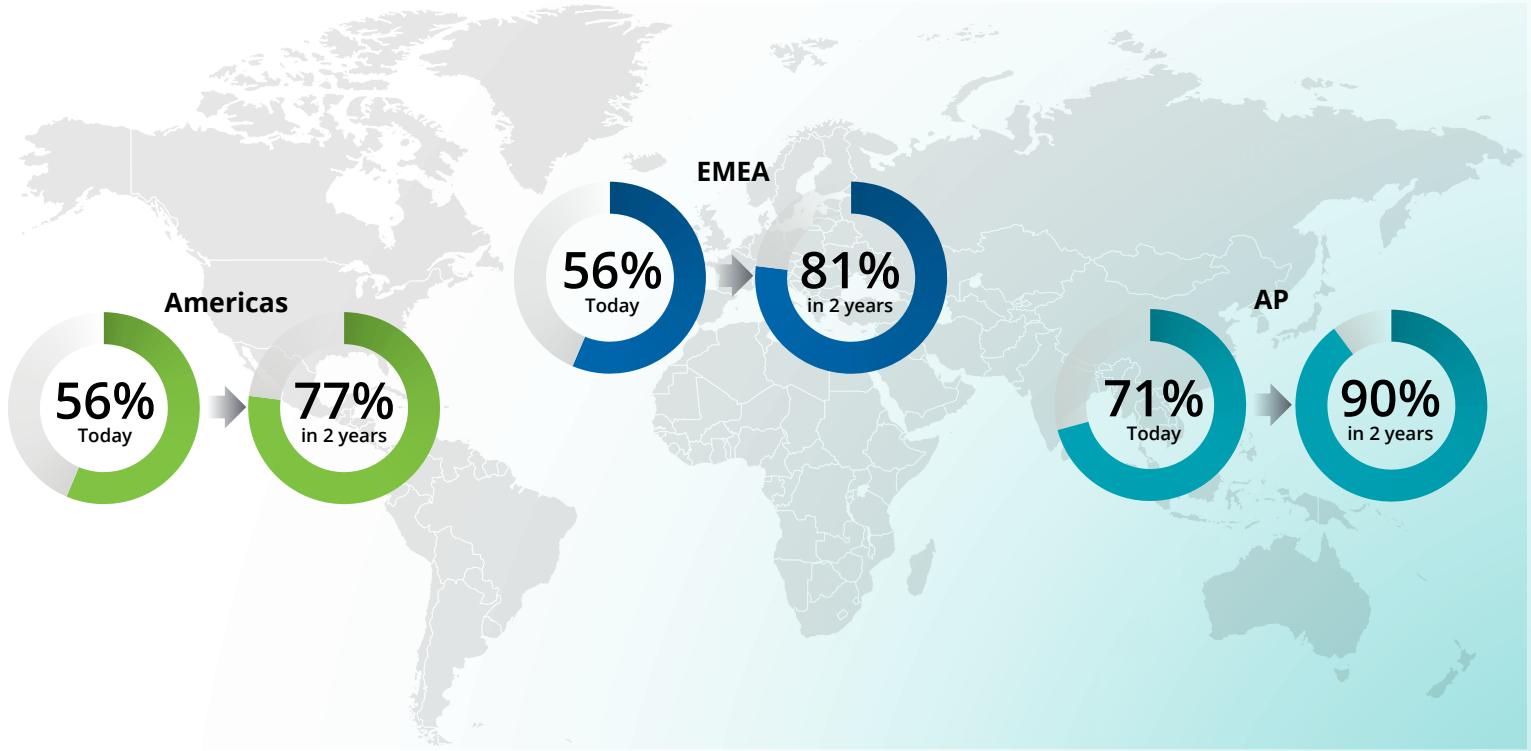
**Figure 9: Extent of physical AI usage** *(percent)*




Question: To what extent is your organization currently utilizing physical AI (e.g., robotics, automated machinery) in its operations?
N=3,235

Survey results indicate that organizations in Asia Pacific (AP) are leading in early implementation of physical AI. 71% of AP respondents report at least minimal use of physical AI, compared with 56% in both the Americas and EMEA. AP not only has the broadest adoption, but also the highest share of organizations reporting moderate or greater usage (20% in AP vs. 17% in the Americas and 18% in EMEA). Looking ahead, 90% of AP respondents expect at least minimal use of physical AI, compared with 77% in the Americas and 81% in EMEA. This is meaningful projected growth. Yet, physical AI's expected adoption curve is notably slower than software-based agentic AI, which is expected to jump from 23% to 74% in the same time frame. The difference reflects the inherent challenges of physical deployment: higher costs and capital requirements, longer development cycles, stricter safety regulations, and the need for specialized hardware and maintenance.

**Figure 10: Extent of physical AI usage** *(percent)*




Question: To what extent is your organization currently utilizing physical AI (e.g., robotics, automated machinery) in its operations?
Americas N=3,235, EMEA N=1,170, AP N=475

## Controlled environments are leading the way

Physical AI applications span a wide range of industrial and commercial settings. For example, one company we interviewed is automating package sorting and routing while granting warehouse robots more autonomy to decide where and how to store items to maximize floorspace. Other common use cases include collaborative robots (cobots) on assembly lines, inspection drones with automated response capabilities, robotic picking arms, and autonomous forklifts. Adoption is especially advanced in manufacturing, logistics, and defense, where robotics, autonomous vehicles, and drones are already reshaping operations.



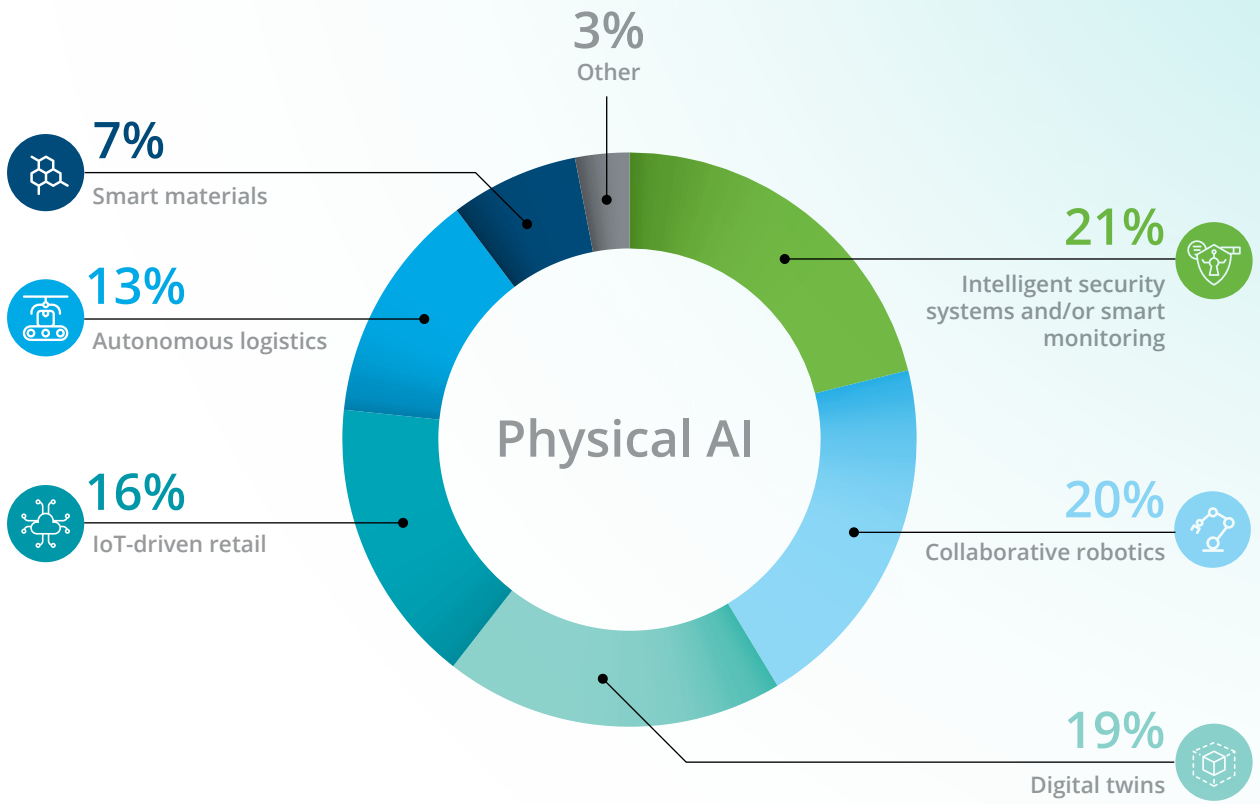
A key factor in early adoption is environmental control. Physical AI use cases that take place in controlled domains such as factories and warehouses tend to progress much faster than use cases in open, real-world environments, where the challenges and risks are far more complex and unpredictable.

## Types of physical AI with the greatest expected impact

As physical AI gains broader adoption, certain types are expected to have a bigger long-term impact than others: intelligent security systems and/or smart monitoring (21%); collaborative robotics (20%); and digital twins (19%) (figure 11).

Interviewed AI leaders shared that smart monitors and digital twins are already significantly transforming operational processes. For example, companies are using 3D mapping of stores to support both interior design and virtual reality training. These detailed digital replicas allow design teams to create customized environments based on actual store scans and also provide realistic training experiences for both customer-facing and retail employees.

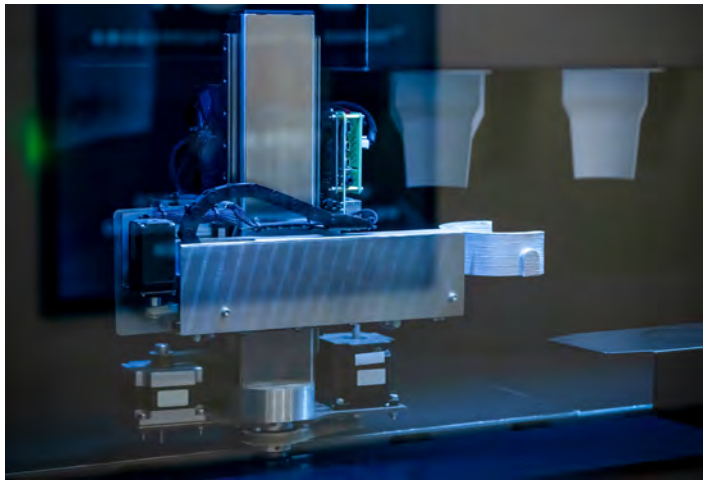
**Figure 11: Types of physical AI expected to have the greatest impact*** *(percent)*




Question: Which area of physical AI do you believe will have the greatest impact on your industry?
* Figures may not sum to 100% due to rounding.
N=3,235

In the restaurant business, AI technologies are being used for targeted marketing and automated inventory management. For example, computer vision enables the automatic tracking of food and beverage items throughout the entire restaurant workflow—from order to delivery—optimizing inventory control and freeing employees from repetitive tasks.

To help ensure public acceptance and trust of these solutions, companies should focus on making them secure, interoperable, and resilient against disruptions and cyberthreats.[3] Unlike software AI that operates in digital environments, physical AI systems interact with people, equipment, and infrastructure in ways that can pose safety risks if they malfunction or are compromised. This raises the stakes for testing, certification, and ongoing monitoring.



Companies must also contend with complex regulatory environments that vary by industry and geography. Physical AI systems often require approval from safety regulators, compliance with industry-specific standards, and adherence to liability frameworks that don't always account for autonomous systems. These requirements add time and cost to deployment but are essential for responsible adoption.

## Accounting for the full cost of physical AI

In our survey, cost was cited most often as a key barrier to physical AI deployment. When evaluating business cases for physical AI, decision-makers should account for total cost of ownership and not just initial equipment costs. The full picture includes facility retrofits to accommodate new equipment, sensors and robots themselves, integration with existing systems and workflows, maintenance and spare parts, and potential downtime during implementation and troubleshooting.

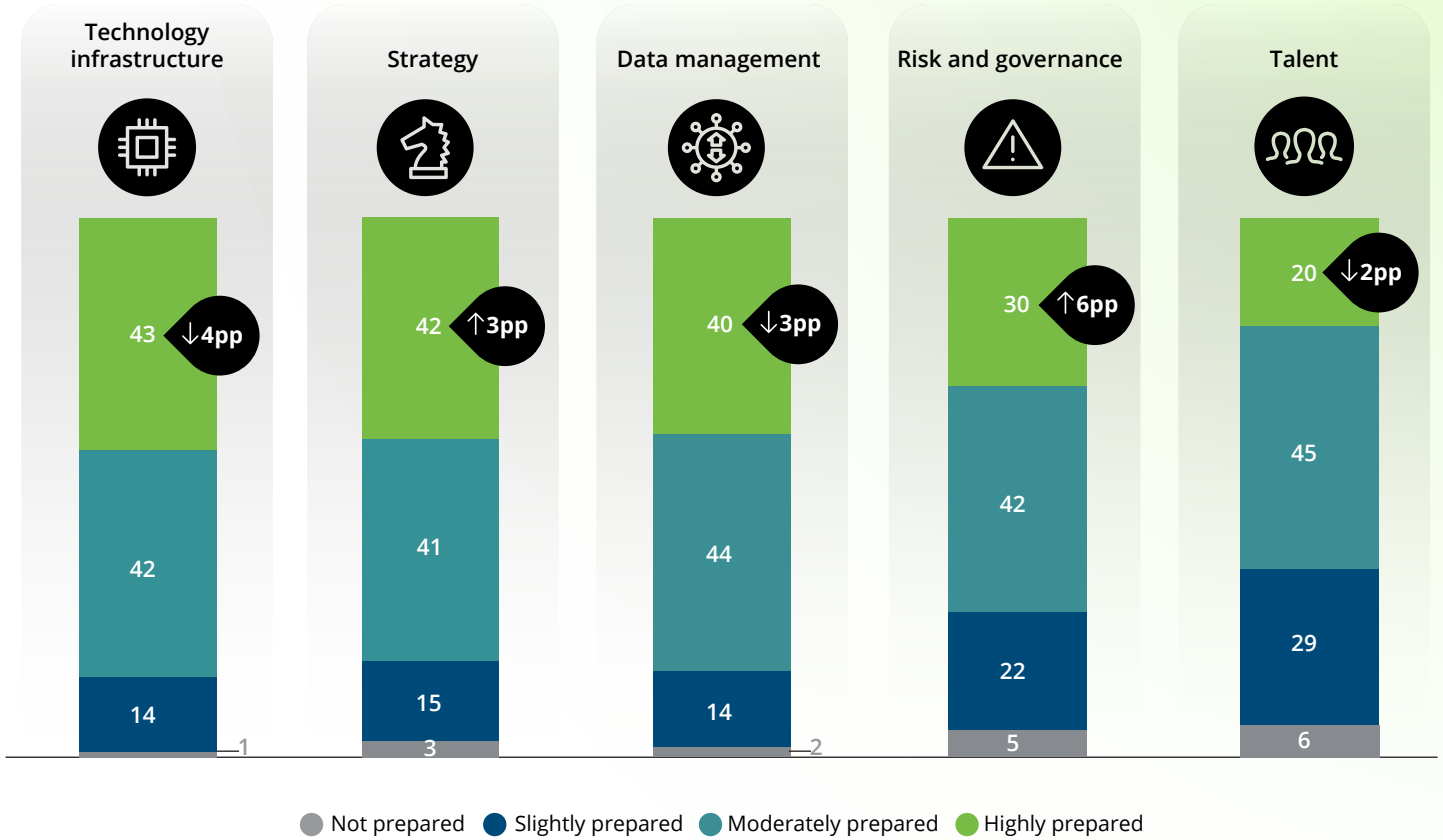


These costs can significantly exceed the initial investment in AI models and software. A warehouse automation project might require hundreds of thousands of dollars in AI development but millions of dollars in physical infrastructure, robotic systems, and facility modifications. Companies that underestimate these costs risk project delays or abandonment partway through implementation.

# Leaders feel more strategically ready for AI than operationally ready in infrastructure and talent

Despite the rapid evolution of AI beyond GenAI to agentic and physical AI, 42% of companies believe their strategy is highly prepared for AI adoption and 30% say the same about risk and governance, both increasing since last year's report (+3 and +6 percentage points, respectively) (figure 12). These areas have likely advanced more quickly as they depend primarily on executive decision-making and policy development.

**Figure 12: Level of preparedness for AI adoption*** *(percent)*




Question: For each, rate your organization's level of preparedness with respect to broadly adopting AI tools/applications.
* Figures may not sum to 100% due to rounding.
N=3,235
Percentage point change vs last year

Meanwhile, perceptions of high preparedness have shifted down compared with last year for technical infrastructure (43%), data management (40%) and talent (20%), revealing the persistent challenge of modernizing systems and skills at the speed of innovation. In fact, most respondents believe that resolving the key challenges for their organization's priority AI initiatives will take more than a year—far too long in today's fast-moving, hyper-competitive marketplace.

As the head of AI strategy at a major European bank explained: "Many organizations prepared for an AI future by building infrastructure and governance for traditional AI models. With LLMs, those efforts were upended. Suddenly, there was a new capability unlike previous AI. Now, traditional AI use cases—training models from scratch, custom interfaces—have diminished. Nearly 80% to 90% of new use cases are generative AI. So yes, companies prepared, but for a different future. GenAI needs a new set of capabilities."

**"Nearly 80–90% of new use cases are generative AI. So yes, companies prepared, but for a different future. GenAI needs a new set of capabilities."**



# Tapping into AI's full potential

The research is telling: AI's transformational potential is real, but capturing it requires far more than just technology investments. Organizations should treat AI as foundational. The most successful won't be those with the most AI projects or the biggest budgets, but those who build AI into the foundation of how they operate, compete, and grow.

**Here are six key focus areas to help your business capture AI's untapped edge:**

## Close the gap between access and activation

Most organizations have deployed AI tools, but far fewer have achieved meaningful usage. The gap between availability and adoption is now the primary barrier to value. Successful companies focus on activation, not just access.

High-performing implementations start with empowered employees who experiment, share early wins, and become internal champions. Top-down directives alone rarely drive meaningful change. Grassroots adoption supported by senior sponsorship creates momentum and helps ensure solutions align with real workflows.

Activation requires early attention to practical constraints: system integration, data permissions, and operational reliability. For organizations applying AI not only to digital processes but also to physical systems—such as robotics, IoT devices, or machinery—early planning for these operational realities is especially critical. Organizations that design for deployment from the outset, rather than treating scale as an afterthought, see far higher adoption. Hands-on, role-specific training and visible executive advocacy materially shift employee behavior. Leaders that treat pilots as stepping stones to production, not isolated experiments, are likely to achieve faster and more durable impact.

## Unlock human advantage by redesigning work around AI

AI is reshaping work at every level. While most organizations currently focus on personal productivity, leaders are rebuilding processes, roles, and career paths around expanded AI capabilities.

The most successful organizations reimagine jobs to seamlessly combine human strengths and AI capabilities, ensuring both aspects are used to their fullest potential. New roles—AI operations managers, human-AI interaction specialists, quality stewards, and others—signal a deeper shift: AI is now a structural component of how work is organized. Advanced organizations streamline workflows that AI can execute end-to-end, while humans focus on judgment, exception handling, and strategic oversight. The goal isn't to replace humans or merely assist them, but to create complementary working relationships between humans and AI, in which the combined output exceeds what either could achieve alone.

Organizational structures are beginning to flatten as AI absorbs routine execution tasks. Some companies are merging technology and people-leadership functions to ensure that systems and workforce design evolve together. The pace varies by industry, but the direction is consistent: Roles, skills, and career paths should be rebuilt, not simply adjusted. Organizations should take an AI-native approach and redesign work holistically rather than layering AI onto legacy processes.



## Build governance before you scale and make it everyone's role

Governance is no longer a compliance exercise; it's the mechanism that enables rapid, confident scaling. Enterprises where senior leadership actively shapes AI governance achieve significantly greater business value than those delegating the work to technical teams alone. True governance makes oversight everyone's role, embedding it into performance rubrics so that as AI handles more tasks, humans take on active oversight. This shared responsibility empowers employees to help identify challenges and guide safe, trusted AI use.

Effective governance integrates with existing risk and oversight structures, not parallel "shadow" functions. It focuses on identifying high-risk applications, enforcing responsible design practices, and ensuring independent validation where appropriate. Leading organizations proactively monitor evolving legal requirements and build systems that can demonstrate safety, fairness, and compliance.

Autonomous systems heighten needs for data and cybersecurity governance.[4] Organizations need to define where humans should remain in control, how automated decisions and data use are audited, and which records of system behavior should be retained. Cross-functional teams—technology, legal, compliance, and business—establish governance frameworks early so that scale does not outpace control. At the same time, governance should be calibrated to balance risk management with innovation, ensuring that oversight enables experimentation rather than constraining it. The objective is not to add bureaucracy but to create clear, adaptive guardrails that allow responsible progress at speed.

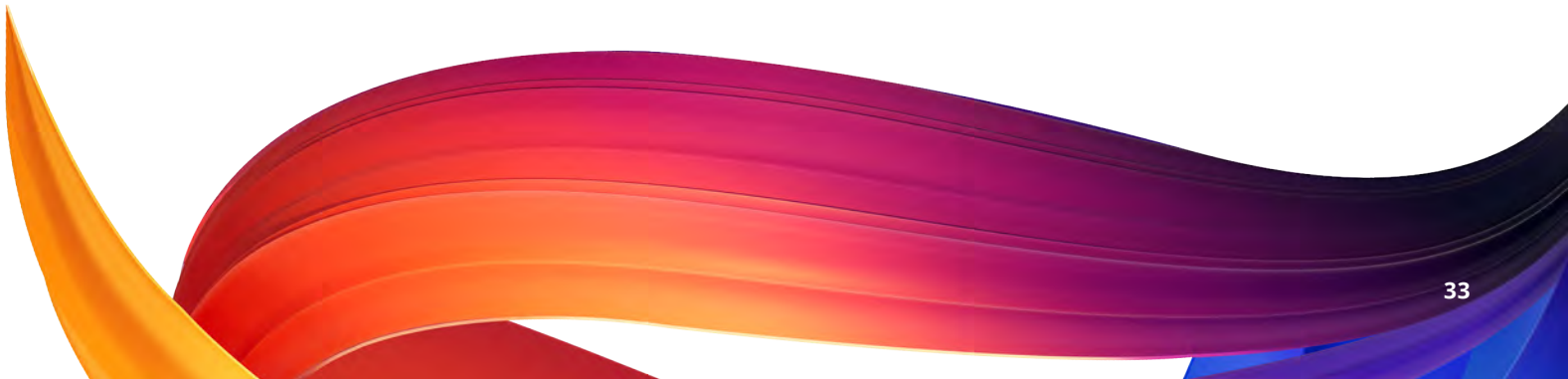


## Address sovereign AI requirements with focus and discipline

As national governments accelerate efforts to establish sovereign AI capabilities, enterprises will likely navigate increasingly complex expectations around data control, model transparency, compliance, and localization. Sovereign AI is no longer limited to the public sector; it is reshaping requirements for every organization handling sensitive data or operating across jurisdictions. At the same time, compute strategy becomes a core component, requiring careful evaluation of both data residency and processing locations (e.g., in cloud, on-prem, hybrid, or edge environments) to remain aligned with evolving regulations and performance needs.

Leading companies take a focused approach: assessing which data and workloads must remain within national or regional boundaries, determining where local model hosting is mandatory, and clarifying how transparency, auditability, and documentation standards differ across markets. They establish clear policies for data residency, model retraining, and cross-border flows, supported by infrastructure capable of meeting multiple regulatory regimes simultaneously.

Enterprises that ignore sovereign AI constraints will face escalating operational disruption, higher compliance risk, and restricted access to key markets. Those that proactively engage build strategic advantage: They can reduce regulatory uncertainty, enhance customer trust, and position themselves as preferred partners in industries where sovereignty concerns dominate. Sovereign AI readiness is now a core element of enterprise resilience and global competitiveness, not a specialized compliance task.

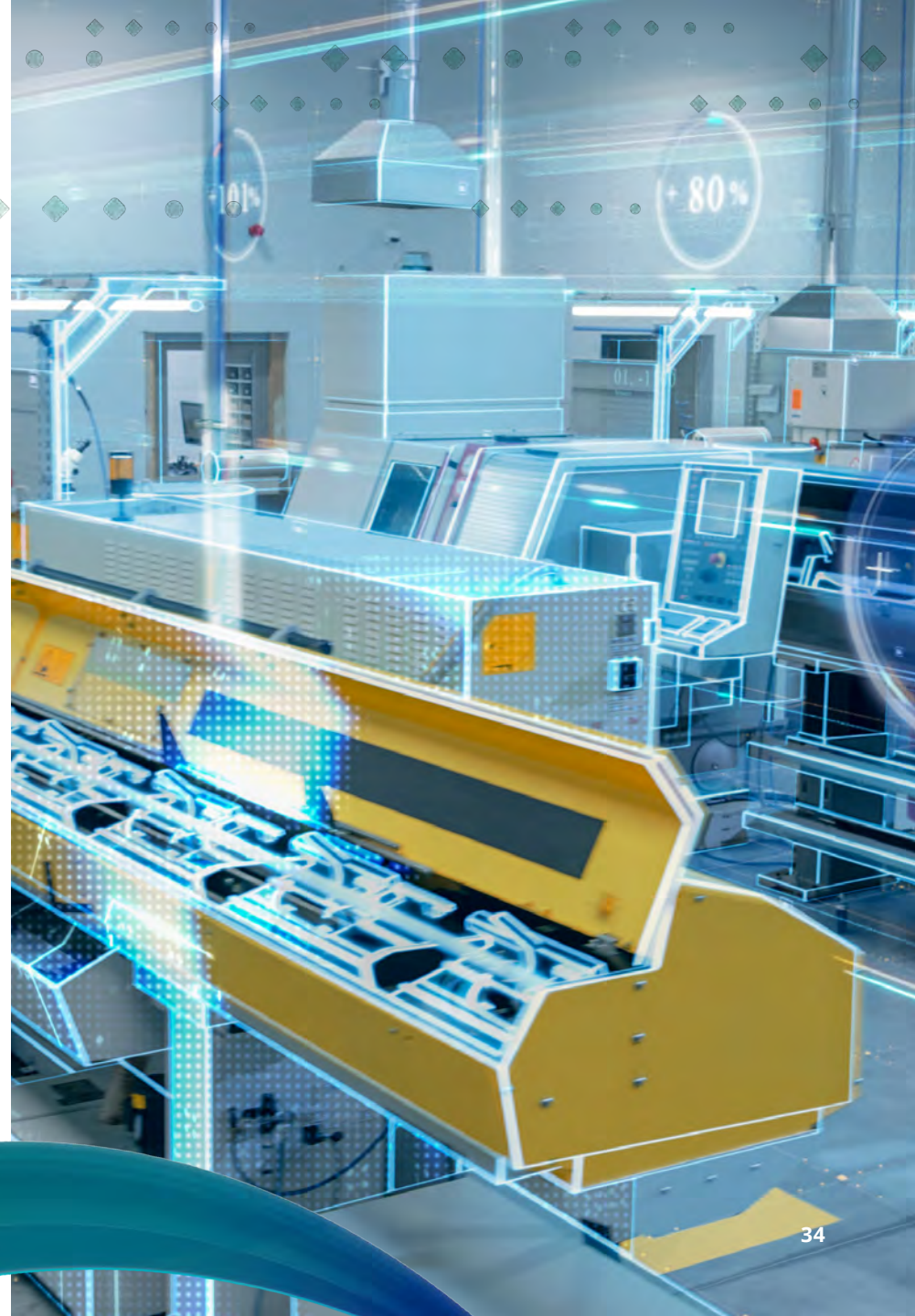


## Build a "living" technology and data infrastructure for tomorrow's AI

Legacy data and infrastructure architectures cannot power real-time, autonomous AI. As AI capabilities extend beyond software into devices, machinery, and edge locations, organizations need to evaluate if their technology foundations are ready to support potential physical AI deployments. Modernization should create a living AI backbone: an organization-wide, real-time system that adapts dynamically to business and regulatory change, elevating infrastructure from IT initiative to strategic capability.

Leaders are enabling modular, cloud-native platforms that securely connect, govern, and integrate all data types, fostering rapid experimentation and seamless scaling. They break down silos with domain-owned data products and embed privacy, sovereignty, and security-by-design, while enforcing enterprise standards for quality, interoperability, and lineage. This balanced approach delivers decentralized innovation supported by centralized control.

A unified, trusted data strategy is indispensable. Poor or fragmented data compounds risk and undermines every AI initiative. Forward-thinking organizations converge operational, experiential, and external data flows and invest in evolving platforms that anticipate the needs of emerging AI.[5] Infrastructure determines enterprise velocity; those that modernize early will likely accelerate while others remain constrained.





## Pursue strategic reinvention, not incremental efficiency

A widening performance divide separates companies treating AI as core to strategy from those viewing it as a cost-saving tool. Leading organizations invest heavily in using AI to reshape operations and create new revenue streams, resisting the pressure to chase every trending technology in favor of initiatives that genuinely advance strategic goals and deliver real value.

These organizations pursue growth across multiple horizons: strengthening current operations, expanding into adjacent markets, and building entirely new businesses enabled by AI. They rethink their organizations from the ground up and imagine how to build without legacy constraints, rather than digitizing old processes. This extends to reimagining business models and adapting to emerging trends like sovereign AI. This intentional reinvention is one of the strongest predictors of achieving outsized returns.

Autonomous AI systems are accelerating this shift. In knowledge-intensive industries, they can absorb substantial routine work, enabling people to focus on higher-order activities. High performers are reorganizing around systems that perceive context, make decisions, and act independently, balancing bold transformation with operational continuity. They move at a pace suited to their organization's readiness, making thoughtful trade-offs and fostering informed decision-making grounded in evidence rather than hype. The strategic opportunity is discovering new sources of value that competitors cannot easily replicate.

With developments in agentic, physical, and sovereign AI rapidly expanding the boundaries of what's possible, companies today are at the edge of tapping into AI's full potential. Whether it's figuring out how to capitalize on the latest cutting-edge innovations, making the leap from pilots to large-scale deployment, or using AI to create an enduring competitive advantage, enterprises around the world are on the edge of transforming themselves with AI. The challenge now is activation: bridging the gap from tool access to meaningful adoption, moving beyond experimentation to operationalizing AI at scale, embedding AI into core business processes—transforming technology potential into enterprise value.

# Acknowledgements

## Authors

### Business leadership




**Jim Rowan**
**US Head of AI**
**Deloitte Consulting LLP**
jimrowan@deloitte.com



**Beena Ammanath**
**Executive Director**
**Global Deloitte AI Institute**
Deloitte LLP
bammanath@deloitte.com



**Nitin Mittal**
**Principal**
**Global AI Leader**
Deloitte Consulting LLP
nmittal@deloitte.com



**Costi Perricos**
**Global GenAI**
**Business Leader**
Deloitte LLP
cperricos@deloitte.co.uk


### Global AI subject matter expert network


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## Technology and research team



**Vivek Kulkarni**
**Managing Director**
**US AI Transformation Leader**
Deloitte Services LP
vivkulkarni@deloitte.com



**Lisa Hohener**
**Senior Manager**
**Deloitte AI Institute**
Deloitte Consulting LLP
lhohener@deloitte.com



**Caroline Ritter**
**Manager**
**US AI Transformation**
Deloitte Services LP
carritter@deloitte.com

## Additional contributors

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# Continue the conversation

## Global Deloitte AI Institute leaders




**Wessel Oosthuizen**
**Africa**
weoosthuizen@deloitte.co.za



**Jan Hejtmánek**
**Central Europe**
jhejtmanek@deloittece.com



**Alfredo Maria Garibaldi**
**Italy**
agaribaldi@deloitte.it



**Carlos Labanda**
**Spanish-Latin America**
clabanda@deloitte.com



**Chris Lewin**
**Asia Pacific**
chrislewin@deloitte.com



**Martín Cabrera**
**Chile**
mcabreraa@deloitte.com



**Tomotake Kozu**
**Japan**
tomotake.kozu@tohmatsu.co.jp



**Tomas Meca Figueras**
**Spain**
tomecafigueras@deloitte.es



**Dr. Elea Wurth**
**Australia**
ewurth@deloitte.com.au



**Roman Fan**
**China**
rfan@deloitte.com.cn



**Nicolas Griedlich**
**Luxembourg**
ngriedlich@deloitte.lu



**Sulabh Soral**
**United Kingdom**
ssoral@deloitte.co.uk



**Geert Hallemeesch**
**Belgium**
ghallemeesch@deloitte.com



**Richard Eudes**
**France**
reudes@deloitte.fr



**Sultanbek Khunkaev**
**Middle East**
sukhunkaev@deloitte.com



**Jefferson Denti**
**Brazil**
jedenti@deloitte.com



**Björn Bringmann**
**Germany**
bbringmann@deloitte.de



**Naser Bakhshi**
**Netherlands**
NBakhshi@deloitte.nl



**Audrey Ancion**
**Canada**
aancion@deloitte.ca



**Prashanth Kaddi**
**India**
kaddip@deloitte.com



**Tiago Durão**
**Portugal**
tdurao@deloitte.pt

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**Let's explore the future of AI together. →**



# Methodology

To obtain a global view of how AI is being adopted by organizations on the leading edge of AI, Deloitte surveyed 3,235 leaders between August and September 2025. Respondents were senior leaders in their organizations and included board and C-suite members, and those at the president, vice president, and director levels. The survey sample was split equally between IT and line-of-business leaders. Twenty-four countries were represented: United States (n=1,200), Canada (n=175) , Brazil (n=115,) Mexico (n=100), United Kingdom (n=220), Germany (n=170), France (n=150), Netherlands (n=50), Italy (n=75), Spain (n=100), Austria (n=50), Portugal (n=15), Poland (n=40), Czechia (n=30), Saudi Arabia (n=70), United Arab Emirates (n=130), Egypt (n=45), Kuwait (n=10), Qatar (n=10), Oman (n=5), Japan (n=100), India (n=200), Singapore (n=75), Australia (n=100). All participating organizations have one or more working implementations of AI being used daily. Plus, they have pilots in place to explore AI or have one or more working implementations of being used daily. Respondents were required to meet one of the following criteria with respect to their organization's AI and data science strategy, investments, implementation approach, and value measurement: influence decision-making, are part of a team that makes decisions, are the final decision-maker, or manage or oversee AI technology implementations. All statistics noted in this report and its graphics are derived from Deloitte's annual survey, conducted between August and September 2025; *The State of AI in the Enterprise* report series. N (Total leader survey responses) = 3,235. The survey data was supplemented with case studies and qualitative findings derived from 15 interviews with executives and AI and data science leaders at large organizations across a range of industries and countries.

## Endnotes


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